100% Deployed & Live

NER-LEWS

AI & Rule-Based Multi-Hazard Early Warning Platform for North East India

GEOGRAPHIC
Focus Area
8 North Eastern States

CRITICAL
Corridors Monitored
NH-6, NH-54, NH-29

ONLINE
Web Application
ner-lews.vercel.app

REST API
Developer Endpoint
https://ner-lews.onrender.com/

The Problem & Proposed Solution

The Regional Problem

Heavy monsoons (>2500mm/yr) trigger sudden and catastrophic slope collapses across the terrain.

These sudden highway blockages completely isolate entire districts for days, endangering lives and severing crucial economic supply chains.

The NER-LEWS Solution

- Transparent MCDA Risk Engine (5-parameter physical hazard calculation)
- Interactive GIS Map (Live animated radar pins via Leaflet)
- Multilingual Avatar Voice (Assamese, Hindi, Bengali, English)
- Citizen Geotagged Field Reports (GPS-captured photo evidence)

Core Differentiators

100% Explainable: Mathematical model with zero black-box ML guesswork.

Zero Cost: No recurring licensing or proprietary software fees.

Field-Ready: Offline-first capability specifically engineered for remote hill areas.

Architecture & Risk Scoring Model

Technical Stack

Engineered for high performance, edge delivery, and accessibility.

- **Frontend:** React 18, Vite, Tailwind CSS, Leaflet.js
- **Backend:** Node.js, Express REST API, Multi-Criteria Decision Analysis (MCDA) Engine
- **Audio Synthesis:** Browser-Native Web Speech API (Client-side execution)
- **Cloud Infrastructure:** Vercel (Edge CDN) + Render (Cloud Backend)

Geotechnical Risk Model

5 Normalized Inputs: 24h Rain (35%), Soil Moisture (30%), Elevation (20%), Humidity (15%), Temp Modifier (+0.5 penalty).

$$\text{Score} = (P \times 0.35) + (SM \times 0.30) + (El \times 0.20) + (H \times 0.15) + T_{mod}$$

- **0.0–3.0:** Low Risk (Routine Monitoring)
- **3.1–6.0:** Moderate Risk (Field Officer Inspection)
- **6.1–8.0:** High Risk (Road Advisory & Alert)
- **8.1–10.0:** Severe Risk (Immediate Evacuation)

System Feasibility & Field Adaptability

Feasibility Strengths

Technical Performance:
Achieves sub-second load times even on degraded 3G/4G networks; fluid responsive design across all mobile and desktop devices.

Financial Viability:
Zero licensing fees; runs efficiently on 100% free-tier open-source cloud infrastructure.

Mass Scalability:
Modular REST API architecture is designed to effortlessly scale and ingest data from 1,000+ IoT telemetry nodes.

Practical Field Solutions

Remote Hill Areas:
Utilizes client-side offline evaluation engine and local seed caching to bypass connectivity blackouts.

Language Barriers:
Deploys native browser voice advisories in regional dialects (Assamese, Hindi, Bengali) for immediate comprehension.

Sensor Gaps:
Incorporates crowdsourced citizen reporting with GPS photo uploads for verified ground-truth data collection.

Real-World Impact & Beneficiaries

Public Safety

Delivers a critical 6 to 24-hour lead time before slope saturation triggers a physical rupture.

Integrated voice alerts ensure the platform remains fully accessible for all citizens, regardless of digital literacy.

Economic Resilience

Protects critical highway freight lifelines (specifically NH-6 and NH-54) by allowing proactive traffic rerouting.

Prevents costly emergency mobilizations and extensive post-disaster infrastructure reconstruction.

Governance & DRR

Equips administrative bodies with a robust, data-backed decision support system.

Aligns directly with the Sendai Framework for Disaster Risk Reduction (Target G for Multi-Hazard Early Warnings).

References & Project Deliverables

Scientific Grounding

The mathematical models and regional thresholds are based on established geotechnical research:

- **Geological Survey of India (GSI)**
  National Landslide Susceptibility Mapping (NLSM) indices.
- **India Meteorological Department (IMD)**
  Eastern Himalayan Rainfall Thresholds for precipitation stress.
- **North Eastern Space Applications Centre (NESAC / ISRO)**
  Early Warning Methodologies for space-based risk mitigation.

Live Deliverables & Attribution

The project is actively maintained and publicly accessible.

Web App: ner-lews.vercel.app

REST API: ner-lews.onrender.com

Development Disclosure

- UI/UX concept and design system structured with Stitch.
- Full-stack engineering and logic implemented in collaboration with Google Antigravity AI.
- Problem definition, mathematical modeling, and platform architecture engineered independently.